

离散事件系统仿真

Discrete-event System Simulation

课程报告

Course Report

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1 Introduction

1.1 Project Description

Traffic inflow is very high to the node and traffic is impacted by tram arrival and departure.

Traffic Signal duration for the lanes are not in sync and phases are not in sequence. Queue length is high in each node except Klosterwuhne.There are Ped Paths present in all the streets. All these above mentioned things when combined provide the traffic node which I have considered for modelling using the simulation software 'Anylogic'. I have then created a conceptual model of the real time system. I have taken records of the traffic and analyzed the data. I have created a program to simulate the real time system using any logic. I then conducted the experiments on the simulated model to check if I can make any further improvements on the system. On the validated simulated model, I have conducted five experiments which have proven to show good results. There Ire also certain limitations to the model because of constraints like time and resources.



Figure 1: Node

1.2 Special Features Of The Node

• Gas Station and a Local Store.
• It's a busy junction with tram lines intersecting the road.
• Bicycle lane on the middle of L¨ubecker Straße.

• Increased traffic due to residential area.

• Ebendorfer Chaussee is one of the entry/exit point to the city.

1.3 Project Quality Criteria

• Implicate five project constraints : time, scope, cost, resources, and risk.
• Accumulate and investigate results from the past experience and implement the modification.
a.Vehicular Operation **b**.Pedestrian Safety **c**.Accident Occurrence

• Comparing real time traffic congestion,to the experimented simulation results.

• Optimize traffic signals time for smooth traffic flow.

• Implementation and maintenance of the optimized solution should be economical.
• Modification to enhance safety of pedestrians, cyclist and other commuters.

• Experiments conducted are in accordance with the traffic rules and are feasible.

1.4 Data Required for the Project

• Vehicle Entry/Exit timing.
• Inter-arrival timing of Vehicles and Pedestrians.

• Traffic Light Cycle.

• Node Features.
• Accident Data.

2 Conceptual Model

2.1 Conceptual Model

A conceptual model is a representation of a system or situation with all its primary parameters. These models are built based on real life scenarios or problems, giving the first hand introduction to what needs to be solved. Petri-Net model is one such conceptual model used in simulation, that defines the states and event transitions of a system. Our Petri-net model describes the functioning of the junction(node which joins the streets L¨ubecker Str., Klosterwhunne Str., Barlaber Str., Ebendorfer Chaussee. Certain assumptions Ire made with justifications to reduce the complexity of real-world scenario. I will also look into the other details like experiments to be modelled, the parameters to be measured as inputs and outputs of the simulation model and the types of events described in the model.

2.2 Assumptions and Reasons

• Vehicles:
In order to simplify the vehicle data I have assumed all vehicles as cars and Large commercial vehicles. These vehicles adhere to the traffic rules and strictly follow the traffic signals as they move from one street to the other. This is done in-order to reduce the complexity of given data.

• Pedestrians and bicyclists:
Pedestrians and bicyclists are considered as a single entity as they have similar behaviour and also have the same signal for crossing. I have also considered them to cross from a single direction.

• Special conditions:
I have considered that the Bicycle lane on L¨ubecker Str. to be idle as the lane is hardly used in the real scenario.

• Gas station:
The gas station and its parking lot are not considered as they have limitations for data acquisition and also this data has no significance on the actual node.

• Traffic signals:

I have considered the traffic signals to function as per the given data, as I have observed deviations from data given to us and the real time scenario.

As it is difficult to remodel the phases as per the current situation, I have considered the phases to function in a sequence from 1-6 and the phases 3 and 6 function only when a tram 1 or 9 arrives respectively.

2.3 Petri-Net Model

A petri-net model is a conceptual model used for simulation. It models all the states and events of a system. This makes it easier to convert the conceptual model into a simulation model. It is generally recommended to use a petri-net model for a small system as the complexity of the model increases with increase in size of the system. This model consists of the agents of the real world represented as dots and named as tokens. The places represented by circles which are the states and events are termed as transitions which are represented as rectangular boxes. There are two types of transitions namely immediate and timed transitions. Immediate transitions are filled boxes and are fired whenever a token is present in the place to which this transition is connected. Timed transitions are hollow boxes which are fired when the
conditions written above them are satisfied. These tokens are created and destroyed in places and this process is triggered by transitions which have specified conditions.

The connections betIen places and transitions are made using lines named as arcs. There are special kind of arcs named as inhibitor arcs which are represented as a line with a small circular head on the end where it is connected to a transition. These inhibitor arcs are used to stop a transition from firing. In this model these inhibitor arcs can be seen near the signals of the streets. The functioning of our model is
detailed in further topics

2.4 Quantities to be measured

• Inter-arrival times:
It is the consecutive time difference betIen agents entering the system. These provide the distribution in which the agents arrive into the system. Modelling the system using these distribution gives the real-time functioning of the sys-
tems. These also provide the probabilities of the directions the arrived agents exit the system.

• Traffic Phase cycle:

It is the time taken for one phase of the traffic signal to change. These provide the queue lengths

of different lanes for which the traffic phase is active(on).

• Delay in vehicle transition:
It is the time taken by an agent to start when a transition is triggered. In our system the time taken for a vehicle to start crossing the lane when the signal turns green. It helps in determining time taken for a vehicle to cross the street

which helps us determine the throughput when a single phase of traffic signal is completed.

• Tram Inter-arrival times:
It is the time difference betIen the arrival of two trams. These are determined by the tram schedule in the data I obtained from the app named INSA (provides tram arrival information for people).

2.5 Quantities used for simulation results

• Throughput:
It is the total number of vehicles moving through the system for a particular amount of time.

This is determined by using the inter-arrival times, traffic phase cycle. It helps in tracking the

amount of vehicles using the node.
• Mean time of Vehicle on the node:
The amount of time taken by a vehicle from arrival to departure of the node.
• Queue Length:
Number of cars accumulated on a street when the signal is Red.

2.6 Types of events in the system

Primary events:
Events which have no conditions for their transitions to occur. In our system, the primary events are
• Arrival and departure of the vehicles.
• Arrival and departure of the pedestrians and bi-cyclists.
• Arrival and departures of the trams.

• Traffic signals.

Secondary events:
Events which have certain conditions for their transitions to occur. In our system, the secondary events are
• Vehicles leaving the node.
• Pedestrians crossing the street.
• Trams leaving the node.

3 Data analysis

Reason behind doing data analysis are to find the input for the simulated model, and to check whether the output matches the output of the real time system. The inter arrival times are calculated for the input and the queue lengths, average time spent by the vehicle in the node are calculated to validate the output.

3.1 Data Collection

For the purpose of data analysis, I took the data from all the streets of the node. i.e L¨ubecker Straße, Ebendorfer Chausse,Klosterwuhne and Barleber Straße.

To get maximum approximation, the data was collected in the evening(16:00 to 17:00)which is the high traffic hour of the node, . I used cameras to record the videos of the traffic for 1 hours form all the streets to find out inter arrival times. I have also taken a note of the traffic signals of all the streets and tram timings so that I can approximate the wave inflow of traffic into the simulation.

3.2 Input and Output Variables

The variables I have considered for input are:
1. inter-arrival times

The variables I have considered for output are:
1. queue length
2. Throughput
3. Average time in the node

3.3 Probability Distribution Functions Estimation

In order to make a right guess of the distribution of cars, I have drawn histograms for the inter arrival times. frequency of cars for particular value of inter-arrival time. This can be observed

in the histograms - figure 6 9,12 and 15. According to the histogram obtained, I have compared with the existing probability distribution functions and assumed that the distribution is log-normal distribution.Repeated the process for all the 4 streets and found that all the distribution as log-normal distribution. In order to check the correctness of our assumption which is log-normal distribution, I have drawn quantile-quantile plot(figure 7,10,13 and 16) Since the plots obtained
in the q-q plots is almost a straight line and the plot passes through origin, the probability distribution function estimation is correct.
I tested our assumed distribution with chi square test and got accepted with 95 percent confidence level(figure 8,11,14 and 17).

Given Data

1. Traffic Phase cycle
2. Accident data
3. Node Dimensions

3.4 Lubecker Straße

3.4.1 Histogram of Lubecker Straße inter-arrival time

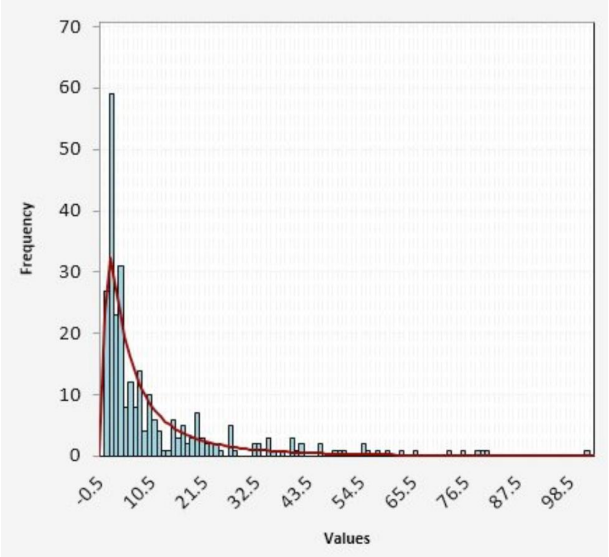


Figure 2: Histogram of Lubecker Straße inter-arrival time

Distribution	Log-normal distribution
Mean	Standard Deviation
1.50	1.26

Table 1: Distribution and parameters

3.4.2 Q-Q plots for Lubecker Straße

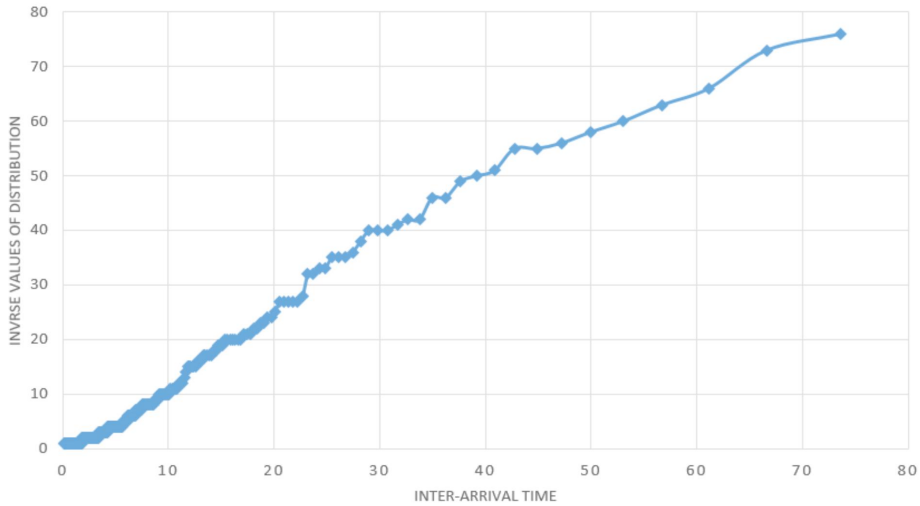


Figure 3: Q-Q plots for Lubecker Straße

3.4.3 Chi square test for Lubecker Straße

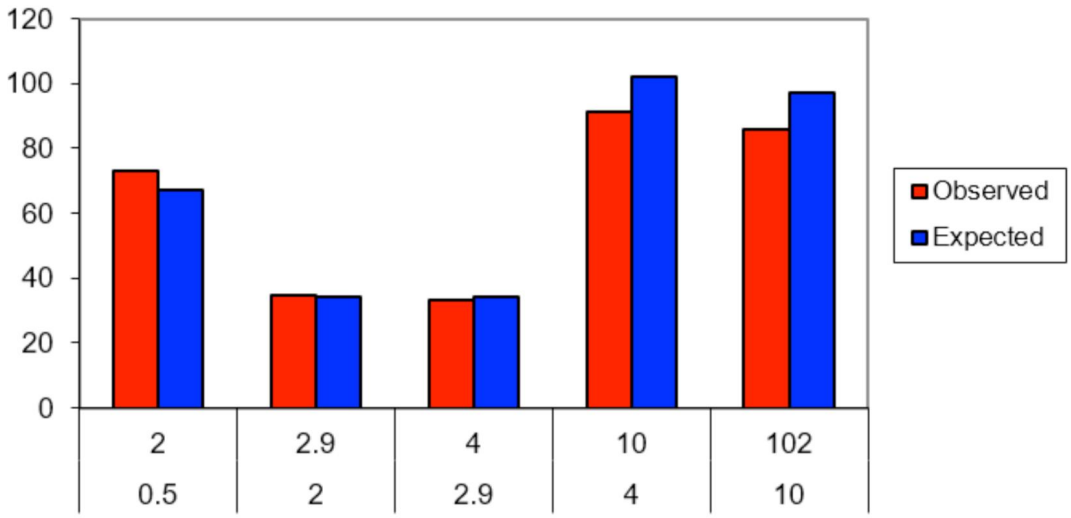


Figure 4: Chi square test for Lubecker Straße

3.4.4 Output Data

• Queue length
• Average time spent

The real-time values for the output variables are mentioned in validation chapter

3.4.5 Difficulties encountered while obtaining the data

•Understanding the process of collecting data and deciding the critical parameters.
•Data collection got postponed due to bad Iather and public holidays.

4 Introduction

The traffic simulation model of the given intersection was built using the simulation software ”Anylogic”.

4.1 Program Structure

The software used for the modelling of the traffic simulation is Anylogic. In Anylogic, when a new model is created they provide us with three main segments. They are Agent, Experiment and Databases.
•In this model, the top level agent is “Main”. This agent is where the intersection is designed and the other agents such as Cars, Pedestrians, Truck and Experiment interact with each other.
•The Experiment segment is used to perform various experiments related to the improvement of safety and improvisation of the node along with the validation of the model.
•The Database is used in this model for providing the arrival of the tram schedule.

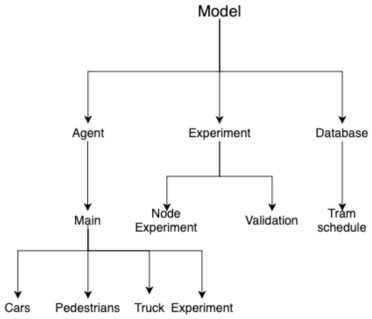


Figure 5: Program structure of Anylogic Model

4.2 Program Concept

The concept of the simulation program involves the modelling of the roads, intersection and behaviour of the agents.

4.2.1 Model Design

The roads in the model are implemented using the “Road” space markup element present in the Road Traffic Library. They are provided with the proper number of lanes according to the real world which can be done in the properties tab of the Road space mark up element.

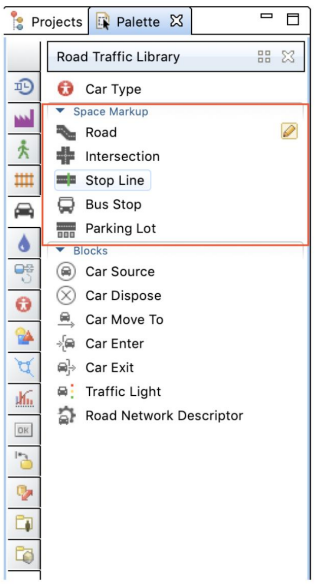


Figure 6: Road Traffic Library - Space Markup elements

Once the roads are all created, the node which is the intersection of all the roads is created using the “Intersection” space markup element in the Road Traffic Library. It inter connects all the roads creating a cross road element in the model similar to the real world. These intersection will provide the connections betIen the roads indication the path for the agents present on the particular lanes.

The tram lines in the real world model was reproduced by creating roads with one forward and one backward lane, which was then connected to the intersection. This was done based on the assumption of considering a Tram as a long car with diff erent properties. Thus, the tram lines Ire modelled as per the real world model.

The traffic lights for the intersection was created by the “Traffic Light” block in the Road Traffic Library. They use stop lines for interaction with the agents. The pedestrian path was built using the “Rectangular wall” space markup element and the “Polygonal Area” in the Pedestrian Library. The source for generation of the pedestrian was created using the “Target Line” from the Pedestrian Library. In the model the pedestrians Ire assumed to move in one direction.

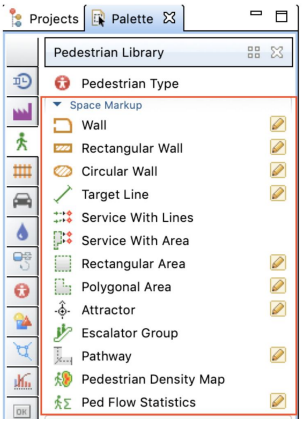


Figure 7: Pedestrian Library - Space Markup elements

4.3 Behaviour of the Agents

4.3.1 Vehicles

The vehicles on the roads Ire classified as Light vehicles (cars) and Heavy vehicles (Trucks). To implement these a Car agent was created and this was used in the Main agent.

The simulation program for Car was built using these four following blocks present in the Road Traffic Library.

- Car Source
- Select Output 5
- Car Move To
- Car Dispose

With the above mentioned blocks the car agents Ire programmed to behave in a similar way as in the real world.

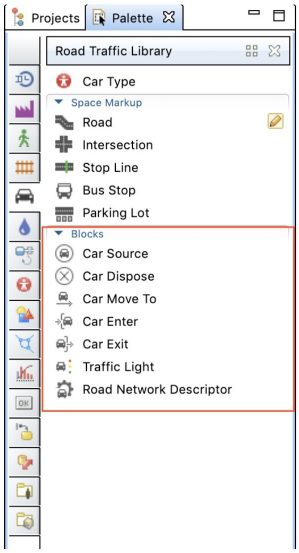


Figure 8: Road Traffic Library - Blocks

Car source is used to generate cars with an inter arrival time distribution specified from the data analysis. Once the car is generated , the select output block is used to select the road to which the car is supposed to move. This was done using the Probability provided in the select output block which determines the input to the car Move to block. When the car agent decides to move on a particular road, they are given as input the Car Move To block which will take the agent to the desired location. Finally the car agent is disposed once it reaches the end of the road.

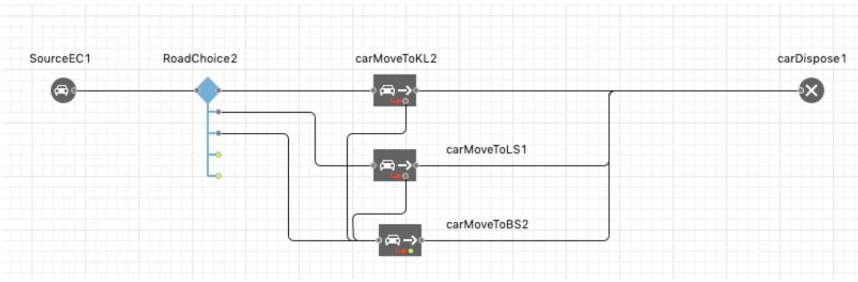


Figure 9: Simulation Program - Car Logic

The same programming logic was used for the Heavy vehicles (Truck).

4.3.2 Pedestrians

The pedestrians Ire modelled using the Pedestrian Library. The behaviour of the pedestrians Ire modelled using the following blocks.

- Ped Source
- Ped Wait Lights
- Ped Move To
- Ped Sink
- Ped Area Descriptor

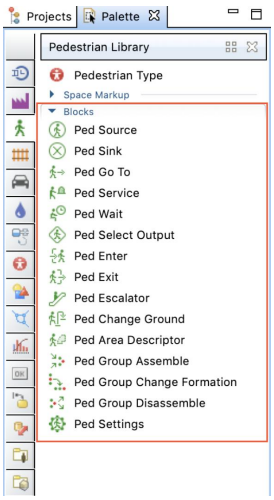


Figure 10: Pedestrian Library - Blocks

The Pedestrians Ire generated using the Ped Source block with respect to the Target Lines. The number of pedestrian generation is provided by the rate at which they will be generated per hour. Once the Pedestrians are generated they should not cross the road at any given time. They should cross the road only during certain phases of Traffic Light. To make this happen the pedestrians have to wait during the other phases. Ped Wait Light block is used to do the exactly same which indicates the pedestrian has to wait in certain area/target line during other phases. When the traffic condition is preferable for the pedestrians, they will move using the Ped Move To block. This will make the pedestrian move to their destination which is their respective target line. Once the pedestrians reached their target line , they will be destroyed using the Ped Dispose block. The Ped Area descriptor is used to control the pedestrians inside the area described in their respective pedestrian area descriptor.

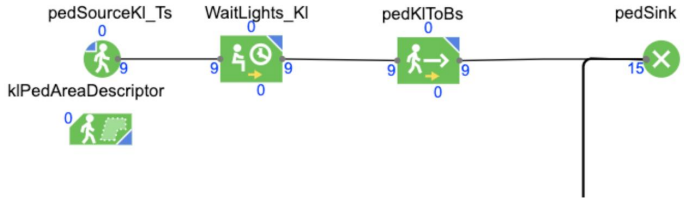


Figure 11: Simulation program - Pedestrian Logic

4.3.3 Trams

The trams are designed exactly the same way as the cars. But the only difference is in the source. The arrival timings of the tram was given in the Tram Schedule Database. This was indicated in the car source block of the trams.

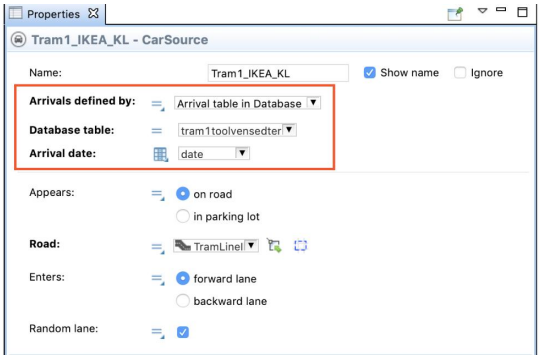


Figure 12: Car Source block modified for Tram Arrival

The following image represents the programming logic used for the Trams.

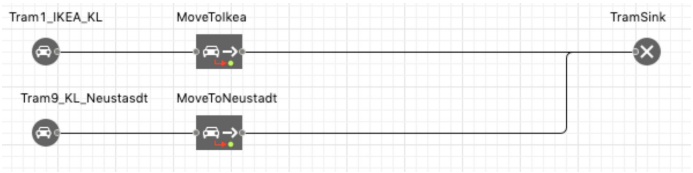


Figure 13: Simulation Program - Tram Logic

4.3.4 Traffic Signal Design

The traffic signal that resembles the real world was designed by considering various phases and the duration of each phase was calculated during the data collection. The traffic phases controlled the respective stop lines in that particular phase.



Figure 14: Traffic light design in Anylogic with respect to stop lines

4.3.5 Statistics

The statistics for the model was provided using the presentation and analysis tools in the Anylogic. The output variable such as the average mean time a car spent on the node, queue lengths of each street.

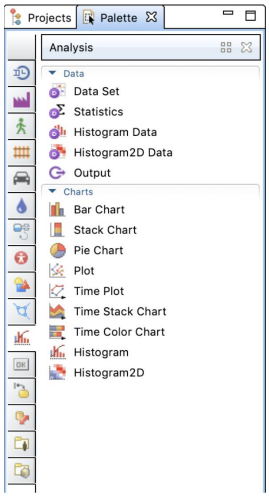


Figure 15: Analysis Library in Anylogic

So using the Bar charts and Histogram charts from the Analysis palette , the following output variables Ire presented as statistics.

- Current Queue Length
- Mean Queue Length
- Mean time spent by Cars on node
- Mean time spent by Trucks on node
- Throughput of the node
- Number of cars in the system at a given time

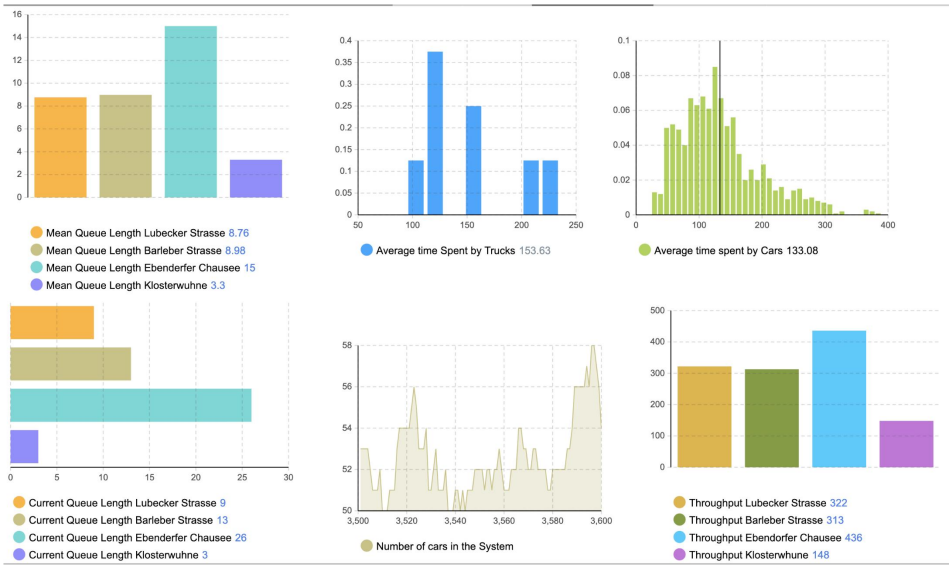


Figure 16: Display of various output variables

4.4 Special Issues

In Lubecker strasse there is a cycle lane in the middle of the lane. So while modelling to resemble this scenario the vehicles should not enter this middle lane. To make this happen, there was new road with two roads was added as in the real world and the traffic was directed in the proper lanes.

4.5 Verification

The model built was verified with the various scenarios such as the congestion in the middle of the intersections, traffic phase cycle, cars waiting for the pedestrians when they are crossing the road, arrival of the trams in the provided timings. These where the major parameters considered for the verification of the model. The verification makes sure that the system is built correctly.

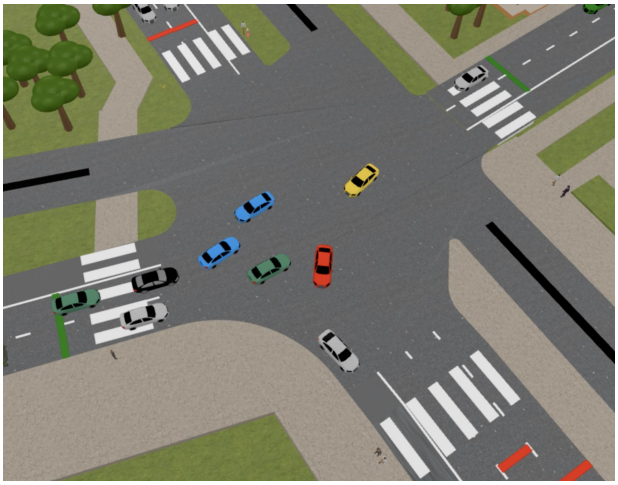


Figure 17: Congestion in the inner node which is similar to the real world

4.6 Difficulties encountered

• Synchronising the tram timings with the traffic phase cycle.

• Designing the phase cycle and calculating the phase.
• Modelling the pedestrians to walk.

The above mentioned issues Ire solved with a lot of effort. And these are considered to be the difficulties encountered during the modelling of the node.

4.7 Experiments Planned

• Removal of the bicycle lane on the L¨ubecker strasse
• Addition of a lane in Ebendorfer Chaussee

• Experimentation with the Traffic phase cycle.

All these above mentioned experiments can be implemented with the present real world model that has been developed with the Anylogic software.

5 Validation

Validation is the process of checking correctness of the model. To determine whether the simulation model is working as intended, Validation is required. Validation statistically proves whether the Simulation model is giving the results within the intended range of accuracy.

5.1 Verification

Simulation model is verified with the Conceptual model, then the behaviour of the Agents in the Simulation model is verified with the behaviour of the Vehicles in the real world. Verification is also done to check whether the simulation program works as intended and it is debugged.

5.2 Validation of Simulation Model

The main purpose of Validation is to check whether the simulation model is representing the real world model at least with respect to the variables of interest. This is done by comparing the results of the model with the data collected from the real world. Figure below describes the Verification and Validation process.

5.3 Quantities of interest

For validation, the Output variables considered are
• Queue Length- Number of cars accumulated in a street when the signal is 'Red'. For this I have considered the Number of Vehicles standing in a street just before the signal turns green.
• Time in Node- Time betIen arrival and departure of a car. For this I have
calculated the time betIen the vehicle generated and the vehicle destroyed.

6 Experiments

6.1 Aim of the Experiments

The main aim of the experiment was to “Improve the traffic safety of Ebendorfer

Chaussee/ Klosterwuhne/ Barleber Straße Node”. For the successful completion of the task, the milestone was divided into several sub-tasks. These are as follows:-

- Analysis of Accident Data.
- Finding the root causes.
- Formulating Experiments.
- Investigating the effect of the experiment on the node.
- Recommending the solutions.

6.2 Analysis of Accident Data

For initiating the experiments,Accident data for the year 2016, 2017 and the year 2018 was provided by “The city council of Magdeburg”.

The primary job was to find out the types of accidents that happened in the past 3 years and also investigate the possible causes. Additionally, to understand the working of the node, several visits to the node was made.

Experiment 1 - Speed Limit

Our atmost priority was to reduce the accident which was caused due to over speeding. To resolve the issue, I decided to apply the speed limit of 30Kmph on all the streets. This will allow the drivers to stop safely in case of an emergency. This will also prevent accidents that may be caused due to reduced traction betIen tires and road during winters or rainy condition.

For the implementation of the “Speed limit experiment” in the Anylogic, a stop line from the road traffic library was assigned to each street and in the property window, the speed limit was assigned as 30kmph. Further, in the property option-Road sign of the main stop line of each street, it was made sure that the end of speed limit was checked.

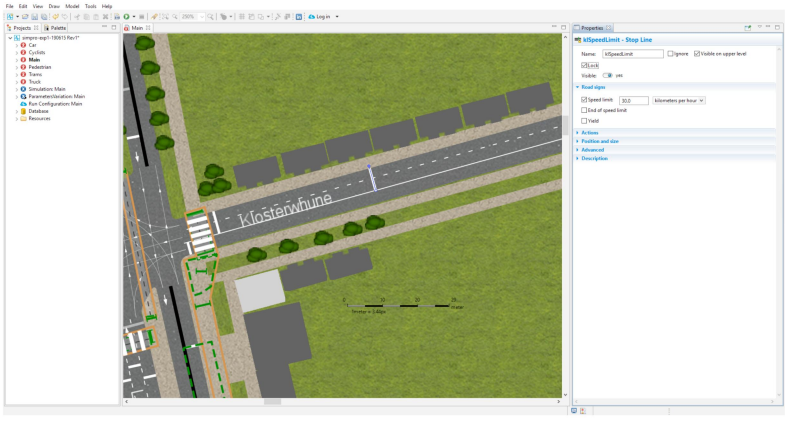


Figure 18: Applying Speed Limit

Experiment 2- Allowing the traffic flow from Klosterwuhne and Ebendorfer Chaussee at different phases

As earlier stated, allowing the vehicles to move from Klosterwuhne and Ebendorfer Chaussee creates a chaotic situation in the node. To solve this issue, I came up with a very simple idea that is to separate the phases for both the streets.

Results of Experiment 1- Speed Limit

By introducing the speed limit in our node, the "Average time spent by cars in the node" increased. Talking about a particular condition of current traffic load, the difference in time is found to be 6.29 seconds. The increased time in the node can be called as a trade-off betIen safety and time.

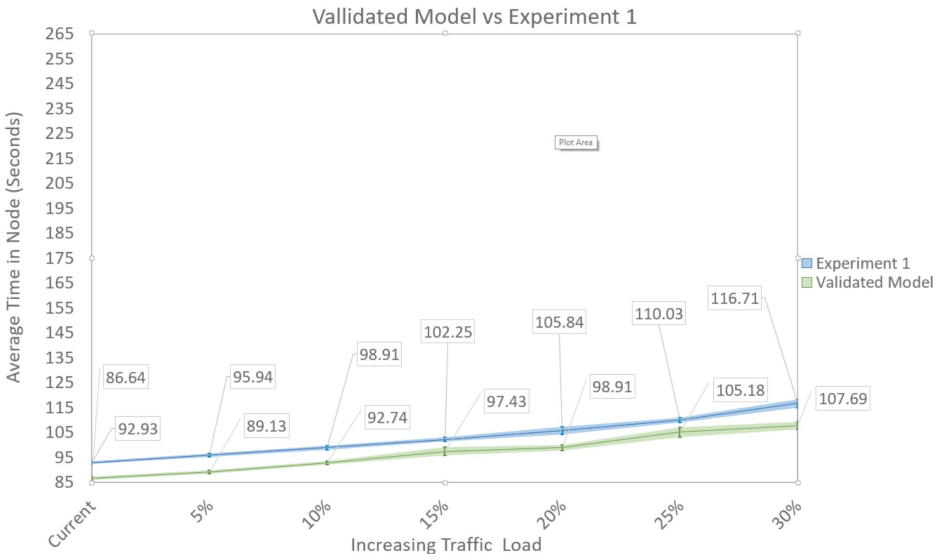


Figure 19: Experiment 1

Results of Experiment 2- Allowing the traffic flow from Kloster wuhne and Ebendorfer Chaussee in different phases

Separating the phase for Ebendorfer Chaussee and Klosterwuhne reduced the congestion in the node but increased the average time spent by vehicles by 5.25 seconds in case of normal traffic load. Although when compared with experiment 1, the difference appears to be small, but as the traffic loads increases, the average time in the node starts increasing more than that in the case of experiment 1.

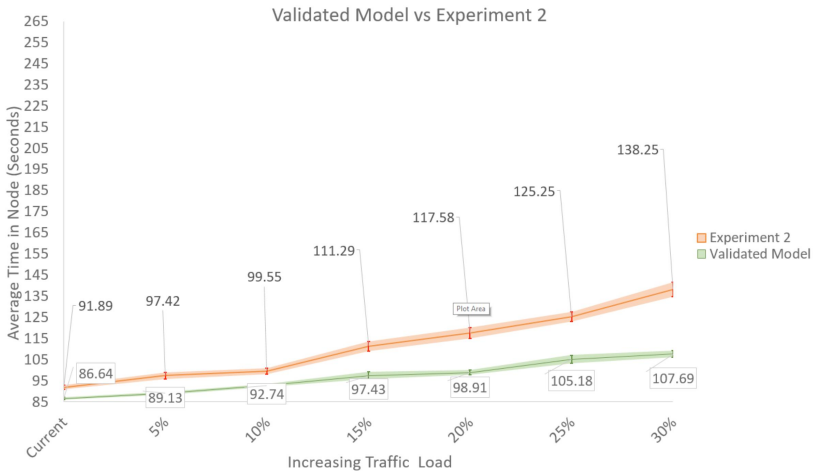


Figure 20: Experiment 2

6.3 Limitations of the Experiment

• As stated earlier, safety is a subjective issue and cannot be measured. Moreover, I are unable to model accidents hence I don't know the uncertainties that can occur in the node.

• The node is tested for increased traffic load by up to 30 percent. In case the

traffic increases more than 30 percent then the functioning of the node is un-

known.

• As stated earlier , I am not able to cover 9 accidents due to unavailability of reasons or maybe the accident happened due to alcohol influence. These accidents may occur even if the recommendations are implemented.

• The experiments done are only to tackle the situation of Over Speeding and Congestion in the node. Influence of other factors like Iather and darkness is not considered.

7. Conclusion

This report presented the development, analysis, and evaluation of a discrete-event traffic simulation model for the Ebendorfer Chaussee–Klosterwuhne–Barleber Straße–Lübecker Straße intersection using the AnyLogic simulation platform. The primary objective of the study was to model real-world traffic behavior at a complex urban node and to investigate potential improvements in traffic safety and operational performance through simulation-based experiments.

A detailed conceptual model was first developed using Petri-net representations to capture the states, events, and interactions among vehicles, pedestrians, trams, and traffic signals. Based on real-world data collected during peak traffic hours, appropriate probability distributions for vehicle inter-arrival times Ire identified and statistically validated using histograms, Q–Q plots,

and chi-square tests. This ensured that the simulation inputs realistically reflected actual traffic conditions at the node.

The simulation model integrated multiple agent types, including light vehicles, heavy vehicles, pedestrians, and trams, each with distinct behavioral logic. Traffic signals Ire modeled according to observed phase cycles, and special infrastructure features such as tram lines and pedestrian crossings Ire incorporated to closely resemble real-world operations. Verification and validation processes confirmed that the model behavior and output metrics—such as queue length, throughput, and average time spent in the node—Ire consistent with observed traffic patterns.

Two main experiments Ire conducted to evaluate potential improvements to traffic safety and congestion. The first experiment introduced a uniform speed limit of 30 km/h across all approaches, which resulted in a modest increase in average travel time but offered a clear safety benefit. The second experiment separated traffic phases for Klosterwuhne and Ebendorfer Chaussee, reducing congestion at the intersection while slightly increasing travel time under normal traffic loads. These results highlight the inherent trade-off betIen safety and efficiency in urban traffic management.

Despite the positive outcomes, the study has certain limitations. Accident occurrences could not be explicitly modeled, and the simulation was tested only under traffic demand increases of up to 30%. Additionally, external factors such as Iather conditions, visibility, and driver behavior variability Ire not considered. Nevertheless, the developed model provides a flexible and extensible framework that can be enhanced with additional data and more advanced behavioral assumptions in future work.

In conclusion, this project demonstrates that discrete-event simulation using AnyLogic is an effective tool for analyzing complex traffic systems and evaluating alternative traffic management strategies. The proposed model can serve as a decision-support framework for urban planners and traffic engineers, enabling data-driven assessment of safety-oriented and congestion-reduction measures before their real-world implementation.